

Guanhua Ji

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EDUCATION

University of Pennsylvania

Aug 2025 - May 2027

M.S.E. in Robotics

Selected coursework in Machine Perception, Learning in Robotics, and Physical Intelligence.

University of California, Los Angeles

Sep 2021 - Jun 2025

B.S. in Computer Science | GPA 3.95/4.00

Dean's Honors List | Upsilon Pi Epsilon member | GRE 330/340

PUBLICATIONS

Guanhua Ji*, Harsha Polavaram*, Lawrence Yunliang Chen*, Sandeep Bajamahal, Zehan Ma, Simeon Adebola, Chenfeng Xu, Ken Goldberg. [OXE-AugE: A Large-Scale Robot Augmentation of OXE for Scaling Cross-Embodiment Policy Learning](#). *ICML 2026 (Spotlight)*.

Guanhua Ji*, Tianyu Li*, Dayoon Suh, Yuqian Zhang, Boyan Zhang, Nadia Figueroa. [Dreaming the Sound of Contact: Leveraging Video and Audio Generation for Zero-Shot Force-Aware Manipulation and Data Generation](#). *Submitted to ICRA 2027*.

(*Equal contribution)

RESEARCH EXPERIENCE

Mechanical Systems Control Lab, UC Berkeley

Research Intern

May 2026 - Present

Advised by Prof. Masayoshi Tomizuka

Haptic Teleoperation for Contact-Rich Imitation Learning (in preparation)

- Identified two mechanisms by which haptic feedback improves demonstration quality for contact-rich imitation learning. Haptic feedback supports force regulation when contact deformation is difficult to see and helps demonstrations encode distinct force-control strategies across task phases.
- Built a teleoperation system connecting a FANUC arm, Touch haptic device, and ATI force/torque sensor. Developed the data-collection pipeline to record robot motion, visual observations, and contact forces for policy training.
- Trained and evaluated flow-matching policies on haptic and non-haptic demonstrations, with a particular focus on hex-key screwing, a challenging manipulation task requiring precise alignment and fine force control.
- Designed controlled experiments to test both mechanisms. Preliminary results showed that haptic feedback improved policy success from 57% to 100% at the vase-writing viewpoint where contact deformation was hardest to observe. Failure analysis and force-history prediction linked clearer task-phase encoding to more reliable hex-key screwing, with policy success of 61.7% versus 13.3% using equal numbers of successful training demonstrations.

Figueroa Robotics Lab, University of Pennsylvania

Research Intern

Oct 2025 - Present

Advised by Prof. Nadia Figueroa

Force-Aware Manipulation from Generated Video and Audio (submitted to ICRA)

- Proposed jointly leveraging generated video and audio for zero-shot force-aware manipulation, addressing the lack of contact-force control in video-only manipulation pipelines.
- Led the development of a pipeline that generates video and synchronized audio from an initial image and task prompt. Reconstructed 3D motion to recover robot trajectories and used generated audio to identify contact intervals and shape desired-force profiles for closed-loop execution.
- Led real-robot experiments across four contact-rich tasks on a Franka Panda, achieving 90% zero-shot success versus 20% for a kinematic-only baseline. Further analysis on wiping and peeling showed that audio-shaped force profiles achieved higher success with lower peak force and force impulse than the best tested constant-force baseline.
- Released 200 force-aware trajectories across four tasks and trained Diffusion Policies on the generated demonstrations, validating the pipeline as a data-generation engine. Policies generalized to variations in object placement and appearance.

AUTOLab, Berkeley AI Research (BAIR), UC Berkeley

Research Assistant

Jan 2025 - Jan 2026

Advised by Prof. Ken Goldberg

Robot Data Augmentation for Cross-Embodiment Policy Learning (ICML 2026 Spotlight)

- Scaled cross-embodiment training data by transforming existing demonstrations into data for diverse robot embodiments while preserving the original tasks and scenes.
- Led the development and open-source release of AugE-Toolkit to retarget demonstrations to up to eight other robot embodiments through segmentation, background inpainting, and simulation-based replay. Released OXE-AugE with over 4.4 million trajectories across nine embodiments, augmenting 16 OXE datasets.
- Ran a systematic scaling study in simulation, showing that greater augmentation diversity improves generalization to unseen robots and robustness to visual changes on the original robot.
- Led physical experiments with OpenVLA-OFT and $\pi 0$ fine-tuned on augmented Bridge data, including $\pi 0$ fine-tuning on 10,000 trajectories. Both outperformed non-augmented baselines across four tasks, reaching average task scores of 75% and 82%, respectively, on an unseen robot–gripper configuration.

TEACHING AND SERVICE**UPE CS & Math Tutor, UCLA****Mar 2023 - Jun 2024**

- Tutored 100+ UCLA students in computer science, discrete mathematics, and algorithms.